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A Words vs. Nonwords Analysis

To further analyze our results from Sec. 3, we conduct a failure analysis, focusing on false negatives (FN) and false positives (FP). Our analysis indicates that the former—valid words misclassified as gibberish—are often rare and complex words, suggesting that the model's internal vocabulary may lack representations for infrequent words. On the other hand, false positives—where gibberish is misclassified as a word—typically involves nonwords that closely resemble valid words, likely due to shared sub-word structures. See Tab. 2 for a few examples.

Original Word	**Predicted Value**	**True Value**	**Status**
_Unitarianism	gibberish	word	FN
_killy	gibberish	word	FN
_quadruropic	word	gibberish	FP
_nonwith	word	gibberish	FP

Table 2: Examples of false negative (FN) and false positive (FP) for the word vs. nonword experiment (Sec. 3).

We further conduct additional experiments to compare the performance of our custom nonword dataset with the linguistically-motivated ARC Nonword Database (Rastle et al., 2002), which contains morphologically plausible nonwords designed to resemble real words to humans. Our results (Fig. 7) indicate a higher differentiability on the ARC dataset compared to our own dataset, which highlights that our nonword creation procedure effectively mitigates potential biases. This result strengthens our conclusions: the last token provides significant cues for distinguishing words from nonwords, particularly in the middle layers of the model after several layers of processing.

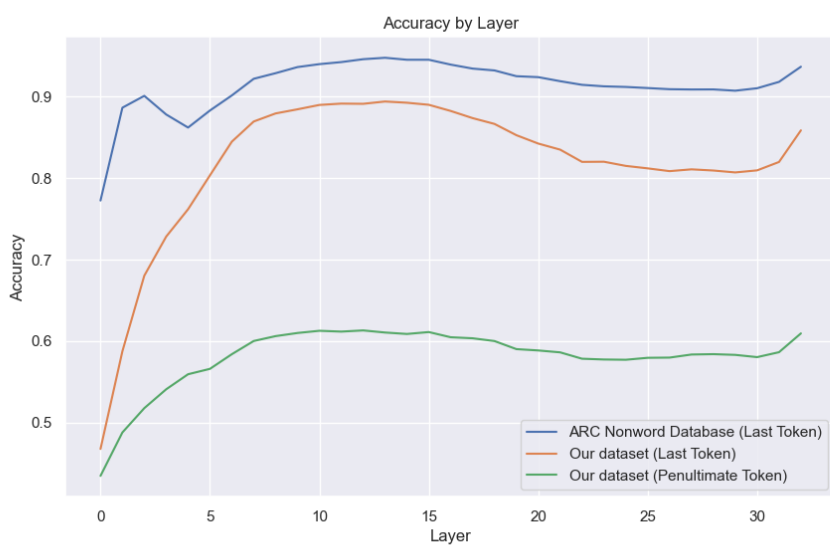

Figure 7: Accuracy comparison by layer for distinguishing words from nonwords. The ARC Nonword Database (blue line) exhibits the highest accuracy across layers, potentially due to its morphologically plausible nonwords. In contrast, our dataset (orange line, last token) achieves slightly lower accuracy, demonstrating the effectiveness of our bias-mitigated nonword generation process. The penultimate token from our dataset (green line) shows significantly lower accuracy, highlighting the importance of the last token in distinguishing words from nonwords.

In another experiment, we evaluate the penultimate token representation against nonword tokens for words of length three or more tokens. For example, in a word like "unhappiness," we use the inner token "h" (e.g., *un-h-appiness*). Unlike the final tokens, penultimate tokens are poorly distinguishable from nonwords, achieving significantly lower accuracies. These results suggest that the model's